

# Tables 1–13 and Appendix Tables

Unifying Fairness Privacy Robustness and Explainability in Federated  
Graph Neural Networks for High Stakes Risk Detection

**Table 1:** Where methods sit across the trust axes. ✓: supported; ∼: partial/adaptable; blank: not addressed. “Graph” = graph-native; “Group fair” = demographic group fairness (not merely client fairness); “Crypto” = validated on a large-scale cryptocurrency graph. <sup>‡</sup>Our DP entry is *not* equivalent to the record-level guarantees of DP-FedAvg, PUFFLE, FDP-Fair or FaVGNN: TrustFedGNN privatises the *released fairness statistic* rather than the whole model update (Section ??), so this column marks the presence of a formal  $(\epsilon, \delta)$  guarantee, not guarantees of equal scope.

| Method                    | Fed. | Graph | Group fair | DP             | Byz. | Explain | Crypto |
|---------------------------|------|-------|------------|----------------|------|---------|--------|
| FairGNN [? ]              |      | ✓     | ✓          |                |      | ∼       |        |
| NIFTY [? ]                |      | ✓     | ✓          |                |      |         |        |
| FairSIN [? ]              |      | ✓     | ✓          |                |      |         |        |
| DAB-GNN [? ]              |      | ✓     | ✓          |                |      |         |        |
| FairFed [? ]              | ✓    |       | ✓          |                |      |         |        |
| q-FedAvg [? ]             | ✓    |       | ∼          |                |      |         |        |
| PUFFLE [? ]               | ✓    |       | ✓          | ✓              |      |         |        |
| FedFACT [? ]              | ✓    |       | ✓          |                |      |         |        |
| DP-FedAvg [? ]            | ✓    |       |            | ✓              |      |         |        |
| Krum/Median [? ? ]        | ✓    |       |            |                | ✓    |         |        |
| F <sup>2</sup> GNN [? ]   | ✓    | ✓     | ✓          |                |      |         |        |
| FairGFL [? ]              | ✓    | ✓     | ∼          | ✓              |      |         |        |
| FaVGNN [? ]               | ✓    | ✓     | ✓          | ✓              |      |         |        |
| FDP-Fair [? ]             | ✓    |       | ✓          | ✓              |      |         |        |
| FedGraph-Fair [? ]        | ✓    | ✓     | ✓          |                |      |         |        |
| PoPETs’25 [? ]            | ✓    |       | ✓          | ✓              |      |         |        |
| <b>TrustFedGNN (ours)</b> | ✓    | ✓     | ✓          | ✓ <sup>‡</sup> | ✓    | ✓       | ✓      |

**Table 2:** Datasets. <sup>†</sup>Elliptic uses an operational early/late temporal subgroup as a documented proxy (not a protected demographic).

| Dataset  | Nodes   | Edges     | Feat. | Sensitive attr.          | Pos. rate | Task             |
|----------|---------|-----------|-------|--------------------------|-----------|------------------|
| German   | 1,000   | 43,484    | 26    | Gender                   | 0.70      | credit risk      |
| Bail     | 18,876  | 623,740   | 17    | Race (WHITE)             | 0.38      | recidivism       |
| Credit   | 30,000  | 2,843,716 | 12    | Age                      | 0.78      | default          |
| Pokec-z  | 67,796  | 1,235,916 | 276   | Region                   | 0.08      | working field    |
| Elliptic | 203,769 | 234,355   | 165   | Time period <sup>†</sup> | 0.02      | illicit (crypto) |

**Table 3:** Paired Wilcoxon signed-rank tests (TrustFedGNN vs. each baseline over shared seeds):  $p$ -values *rounded to two decimals*, \* marks  $p < 0.05$ ; where the running text quotes a  $p$ -value it quotes this rounded value. “–” where fewer than five shared seeds are available (Credit/Elliptic remain at 2–3 seeds by cost, and are run against a restricted baseline set—see Section ??). Note the Wilcoxon floor: at  $n=5$  (Bail, Pokec) the smallest attainable  $p$  is 0.0625, so a uniform 0.06 indicates our method wins on *all five* paired seeds, not a weak effect. On German (10 seeds) TrustFedGNN’s DPD is *not* significantly different from the non-fair backbone or most fair baselines; its AUC is significantly higher than the other (collapsing) privacy methods and lower than the non-private ones. On Bail its AUC beats every baseline on all seeds while its DPD ties the fairest non-collapsed methods; on Pokec it beats the backbone on *both* axes on all seeds.

| Metric | Baseline           | German | Bail | Credit | Pokec-z | Elliptic |
|--------|--------------------|--------|------|--------|---------|----------|
| AUC    | FedAvg-GAT         | 0.02*  | 0.06 | –      | 0.31    | –        |
| AUC    | FairGNN            | 0.16   | 0.06 | –      | 0.06    | –        |
| AUC    | FairSIN            | 0.02*  | 0.06 | –      | 0.19    | –        |
| AUC    | FairFed            | 0.01*  | 0.06 | –      | 0.06    | –        |
| AUC    | F <sup>2</sup> GNN | 0.02*  | 0.06 | –      | 0.06    | –        |
| AUC    | FDP-Fair (2026)    | 0.01*  | 0.06 | –      | 0.06    | –        |
| AUC    | PUFFLE             | 0.00*  | 0.06 | –      | 0.06    | –        |
| AUC    | DP-FedAvg          | 0.01*  | 0.06 | –      | 0.06    | –        |
| DPD    | FedAvg-GAT         | 0.92   | 0.62 | –      | 0.06    | –        |
| DPD    | FairGNN            | 0.49   | 0.31 | –      | 0.06    | –        |
| DPD    | FairSIN            | 0.49   | 0.81 | –      | 0.06    | –        |
| DPD    | FairFed            | 0.01*  | 0.12 | –      | 0.31    | –        |
| DPD    | F <sup>2</sup> GNN | 0.11   | 0.81 | –      | 0.44    | –        |
| DPD    | FDP-Fair (2026)    | 0.00*  | 0.06 | –      | 0.06    | –        |
| DPD    | PUFFLE             | 0.01*  | 0.06 | –      | 0.06    | –        |
| DPD    | DP-FedAvg          | 0.01*  | 0.06 | –      | 0.12    | –        |

**Table 4:** Utility (AUC-ROC, higher is better). Mean  $\pm$  std over seeds: **10** on the small/high-variance German, **5** on Bail and Pokec, and 2–3 on the larger Credit/Elliptic (see Table 3 for paired tests). Best per column in bold. TrustFedGNN is the only method that is fair *and* private (methods above the rule are non-private). \*FaVGNN is a horizontal adaptation of a vertical-FL method. FairGFL/FedGraph-Fair/PUFFLE/FedFACT/PoPETs’25 are 2025–2026 reimplementations with approximated fidelity (Appendix ??).

| Method               | German                              | Bail                                | Credit                              | Pokec-z                             | Elliptic                            |
|----------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| FedAvg-GCN           | 0.715 $\pm$ 0.040                   | 0.963 $\pm$ 0.006                   | 0.762 $\pm$ 0.005                   | 0.761 $\pm$ 0.004                   | –                                   |
| FedAvg-GAT           | 0.708 $\pm$ 0.052                   | 0.972 $\pm$ 0.001                   | 0.765 $\pm$ 0.000                   | 0.774 $\pm$ 0.010                   | <b>0.970 <math>\pm</math> 0.001</b> |
| FairGNN              | 0.668 $\pm$ 0.084                   | 0.906 $\pm$ 0.060                   | 0.727 $\pm$ 0.001                   | 0.703 $\pm$ 0.022                   | –                                   |
| FairSIN              | 0.689 $\pm$ 0.043                   | 0.956 $\pm$ 0.003                   | 0.761 $\pm$ 0.002                   | 0.762 $\pm$ 0.016                   | 0.969 $\pm$ 0.001                   |
| FairFed              | <b>0.715 <math>\pm</math> 0.045</b> | 0.961 $\pm$ 0.004                   | 0.755 $\pm$ 0.000                   | 0.756 $\pm$ 0.013                   | –                                   |
| q-FedAvg             | 0.712 $\pm$ 0.051                   | 0.961 $\pm$ 0.007                   | <b>0.766 <math>\pm</math> 0.002</b> | 0.762 $\pm$ 0.016                   | –                                   |
| F <sup>2</sup> GNN   | 0.697 $\pm$ 0.043                   | 0.971 $\pm$ 0.004                   | 0.753 $\pm$ 0.000                   | 0.765 $\pm$ 0.010                   | –                                   |
| FaVGNN* (2026)       | 0.626 $\pm$ 0.071                   | 0.870 $\pm$ 0.101                   | –                                   | 0.640 $\pm$ 0.118                   | –                                   |
| FDP-Fair (2026)      | 0.534 $\pm$ 0.062                   | 0.552 $\pm$ 0.114                   | –                                   | 0.613 $\pm$ 0.042                   | –                                   |
| FairGFL (2025)       | 0.708 $\pm$ 0.042                   | 0.963 $\pm$ 0.006                   | –                                   | 0.764 $\pm$ 0.005                   | –                                   |
| FedGraph-Fair (2026) | 0.689 $\pm$ 0.049                   | 0.964 $\pm$ 0.007                   | –                                   | 0.759 $\pm$ 0.006                   | –                                   |
| PUFFLE               | 0.513 $\pm$ 0.076                   | 0.606 $\pm$ 0.097                   | –                                   | 0.561 $\pm$ 0.055                   | –                                   |
| FedFACT (2025)       | 0.706 $\pm$ 0.051                   | 0.963 $\pm$ 0.006                   | –                                   | 0.758 $\pm$ 0.005                   | –                                   |
| PoPETs’25 FairFed    | 0.707 $\pm$ 0.041                   | 0.965 $\pm$ 0.005                   | –                                   | 0.761 $\pm$ 0.009                   | –                                   |
| DP-FedAvg            | 0.523 $\pm$ 0.070                   | 0.587 $\pm$ 0.119                   | 0.481 $\pm$ 0.040                   | 0.615 $\pm$ 0.043                   | –                                   |
| TrustFedGNN (no DP)  | 0.634 $\pm$ 0.028                   | <b>0.990 <math>\pm</math> 0.004</b> | 0.752 $\pm$ 0.008                   | 0.791 $\pm$ 0.011                   | 0.943 $\pm$ 0.003                   |
| <b>TrustFedGNN</b>   | 0.628 $\pm$ 0.039                   | 0.988 $\pm$ 0.006                   | 0.747 $\pm$ 0.013                   | 0.789 $\pm$ 0.016                   | 0.944 $\pm$ 0.004                   |
| TrustFedGNN-Robust   | 0.667 $\pm$ 0.018                   | 0.985 $\pm$ 0.005                   | 0.760 $\pm$ 0.003                   | <b>0.791 <math>\pm</math> 0.005</b> | 0.946 $\pm$ 0.003                   |

**Table 5:** Demographic Parity Difference (DPD, lower is fairer).

| Method               | German                              | Bail                                | Credit                              | Pokec-z                             | Elliptic                            |
|----------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| FedAvg-GCN           | $0.075 \pm 0.056$                   | $0.065 \pm 0.006$                   | $0.064 \pm 0.003$                   | $0.033 \pm 0.007$                   | –                                   |
| FedAvg-GAT           | $0.080 \pm 0.060$                   | $0.070 \pm 0.005$                   | $0.065 \pm 0.002$                   | $0.034 \pm 0.007$                   | $0.061 \pm 0.006$                   |
| FairGNN              | $0.088 \pm 0.064$                   | $0.047 \pm 0.028$                   | $0.104 \pm 0.008$                   | $0.039 \pm 0.014$                   | –                                   |
| FairSIN              | $0.067 \pm 0.044$                   | $0.065 \pm 0.007$                   | $0.062 \pm 0.007$                   | $0.032 \pm 0.008$                   | $0.038 \pm 0.011$                   |
| FairFed              | $0.027 \pm 0.021$                   | $0.054 \pm 0.009$                   | <b><math>0.003 \pm 0.003</math></b> | $0.008 \pm 0.006$                   | –                                   |
| q-FedAvg             | $0.085 \pm 0.047$                   | $0.068 \pm 0.005$                   | $0.064 \pm 0.005$                   | $0.032 \pm 0.006$                   | –                                   |
| F <sup>2</sup> GNN   | $0.038 \pm 0.028$                   | $0.064 \pm 0.005$                   | $0.007 \pm 0.001$                   | $0.010 \pm 0.009$                   | –                                   |
| FaVGNN* (2026)       | $0.069 \pm 0.054$                   | $0.049 \pm 0.016$                   | –                                   | $0.027 \pm 0.024$                   | –                                   |
| FDP-Fair (2026)      | <b><math>0.006 \pm 0.005</math></b> | $0.001 \pm 0.000$                   | –                                   | <b><math>0.001 \pm 0.001</math></b> | –                                   |
| FairGFL (2025)       | $0.080 \pm 0.052$                   | $0.068 \pm 0.005$                   | –                                   | $0.033 \pm 0.006$                   | –                                   |
| FedGraph-Fair (2026) | $0.111 \pm 0.043$                   | $0.067 \pm 0.004$                   | –                                   | $0.032 \pm 0.005$                   | –                                   |
| PUFFLE               | $0.011 \pm 0.007$                   | <b><math>0.001 \pm 0.001</math></b> | –                                   | $0.003 \pm 0.002$                   | –                                   |
| FedFACT (2025)       | $0.077 \pm 0.062$                   | $0.021 \pm 0.014$                   | –                                   | $0.024 \pm 0.010$                   | –                                   |
| PoPETs’25 FairFed    | $0.073 \pm 0.052$                   | $0.066 \pm 0.005$                   | –                                   | $0.033 \pm 0.005$                   | –                                   |
| DP-FedAvg            | $0.010 \pm 0.011$                   | $0.001 \pm 0.001$                   | $0.009 \pm 0.005$                   | $0.004 \pm 0.002$                   | –                                   |
| TrustFedGNN (no DP)  | $0.040 \pm 0.035$                   | $0.070 \pm 0.013$                   | $0.015 \pm 0.009$                   | $0.012 \pm 0.009$                   | $0.014 \pm 0.014$                   |
| <b>TrustFedGNN</b>   | $0.079 \pm 0.059$                   | $0.066 \pm 0.013$                   | $0.037 \pm 0.014$                   | $0.013 \pm 0.005$                   | $0.039 \pm 0.008$                   |
| TrustFedGNN-Robust   | $0.081 \pm 0.086$                   | $0.058 \pm 0.008$                   | $0.026 \pm 0.002$                   | $0.014 \pm 0.007$                   | <b><math>0.003 \pm 0.002</math></b> |

**Table 6:** Equal Opportunity Difference (EOD, lower is fairer).

| Method               | German                              | Bail                                | Credit                              | Pokec-z                             | Elliptic                            |
|----------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| FedAvg-GCN           | $0.087 \pm 0.073$                   | $0.021 \pm 0.007$                   | $0.097 \pm 0.028$                   | $0.033 \pm 0.023$                   | –                                   |
| FedAvg-GAT           | $0.081 \pm 0.067$                   | $0.024 \pm 0.008$                   | $0.096 \pm 0.025$                   | $0.023 \pm 0.008$                   | $0.020 \pm 0.010$                   |
| FairGNN              | $0.058 \pm 0.063$                   | $0.017 \pm 0.012$                   | $0.117 \pm 0.011$                   | $0.013 \pm 0.009$                   | –                                   |
| FairSIN              | $0.074 \pm 0.063$                   | $0.025 \pm 0.013$                   | $0.103 \pm 0.028$                   | $0.018 \pm 0.010$                   | <b><math>0.019 \pm 0.012</math></b> |
| FairFed              | <b><math>0.046 \pm 0.024</math></b> | $0.018 \pm 0.010$                   | $0.072 \pm 0.019$                   | $0.032 \pm 0.020$                   | –                                   |
| q-FedAvg             | $0.095 \pm 0.055$                   | $0.024 \pm 0.011$                   | $0.105 \pm 0.033$                   | $0.038 \pm 0.026$                   | –                                   |
| F <sup>2</sup> GNN   | $0.047 \pm 0.038$                   | $0.021 \pm 0.009$                   | $0.073 \pm 0.010$                   | $0.037 \pm 0.022$                   | –                                   |
| FaVGNN* (2026)       | $0.084 \pm 0.088$                   | $0.020 \pm 0.008$                   | –                                   | $0.016 \pm 0.008$                   | –                                   |
| FDP-Fair (2026)      | $0.120 \pm 0.107$                   | $0.006 \pm 0.004$                   | –                                   | $0.014 \pm 0.009$                   | –                                   |
| FairGFL (2025)       | $0.071 \pm 0.049$                   | $0.028 \pm 0.008$                   | –                                   | $0.030 \pm 0.013$                   | –                                   |
| FedGraph-Fair (2026) | $0.090 \pm 0.041$                   | $0.022 \pm 0.009$                   | –                                   | $0.036 \pm 0.024$                   | –                                   |
| PUFFLE               | $0.102 \pm 0.088$                   | $0.006 \pm 0.013$                   | –                                   | $0.023 \pm 0.013$                   | –                                   |
| FedFACT (2025)       | $0.097 \pm 0.077$                   | $0.012 \pm 0.007$                   | –                                   | $0.088 \pm 0.041$                   | –                                   |
| PoPETs'25 FairFed    | $0.072 \pm 0.046$                   | $0.023 \pm 0.008$                   | –                                   | $0.022 \pm 0.020$                   | –                                   |
| DP-FedAvg            | $0.139 \pm 0.090$                   | $0.012 \pm 0.008$                   | $0.108 \pm 0.083$                   | <b><math>0.011 \pm 0.007</math></b> | –                                   |
| TrustFedGNN (no DP)  | $0.061 \pm 0.062$                   | $0.015 \pm 0.015$                   | <b><math>0.065 \pm 0.006</math></b> | $0.024 \pm 0.011$                   | $0.058 \pm 0.022$                   |
| <b>TrustFedGNN</b>   | $0.079 \pm 0.078$                   | $0.009 \pm 0.007$                   | $0.068 \pm 0.012$                   | $0.019 \pm 0.013$                   | $0.075 \pm 0.021$                   |
| TrustFedGNN-Robust   | $0.087 \pm 0.096$                   | <b><math>0.005 \pm 0.004</math></b> | $0.083 \pm 0.022$                   | $0.021 \pm 0.017$                   | $0.079 \pm 0.013$                   |

**Table 7:** Ablation of TrustFedGNN components (no DP). AUC / DPD per dataset. The ablation is a *separate* set of runs from the main comparison (its own matched seed set, so that all four configurations are compared on identical seeds); its numbers are therefore close to, but not identical with, the corresponding rows of Tables 4 and 5, which aggregate the full seed sets (10 on German, 5 on Bail). All four differences lie within one standard deviation of the multi-seed mean—e.g. German DPD 0.045 here versus  $0.040 \pm 0.035$  there, and Bail AUC 0.988 versus  $0.990 \pm 0.004$ —so the comparison within this table is internally consistent even where a cell does not reproduce the main table exactly.

| Config           | Bail  |       | German |       |
|------------------|-------|-------|--------|-------|
|                  | AUC   | DPD   | AUC    | DPD   |
| Base GAT         | 0.972 | 0.072 | 0.709  | 0.050 |
| + BFWA (no FSER) | 0.972 | 0.073 | 0.680  | 0.071 |
| + FSER (no BFWA) | 0.987 | 0.059 | 0.668  | 0.009 |
| + FSER + BFWA    | 0.988 | 0.075 | 0.638  | 0.045 |

**Table 8:** Attribute-inference attack on the released fairness statistic (Bail, real predictions). A worst-case differencing adversary infers a target node’s sensitive attribute. The exact release is a perfect oracle; FTGD’s noise reduces inference to balanced-random (0.500) at every deployed  $\epsilon$ . Targets are sampled in equal numbers from each true group (2,000 each), so 0.500—not the dataset’s 0.53 majority-group prevalence—is the correct chance reference. Each accuracy is a Monte-Carlo estimate over 4,000 target draws (standard error  $\approx 0.008$ ); no FTGD row differs significantly from 0.500 (two-sided  $p \geq 0.10$ ).

| Release                                | $\epsilon$ | Noise mult. $z$ | Inference acc. |
|----------------------------------------|------------|-----------------|----------------|
| Exact ( $s$ -statistic, no DP)         | $\infty$   | 0               | 1.000          |
| FTGD                                   | 16         | 3.6             | 0.502          |
| FTGD                                   | 8          | 6.4             | 0.510          |
| FTGD                                   | 4          | 11.6            | 0.506          |
| FTGD                                   | 2          | 21.5            | 0.513          |
| FTGD                                   | 1          | 40.5            | 0.507          |
| FTGD                                   | 0.5        | 76.7            | 0.503          |
| <i>Random guess (balanced targets)</i> | —          | —               | 0.500          |



**Table 9:** Utility *and* fairness retained under attack (2/10 Byzantine clients, Bail, single seed). AUC (higher = more robust), DPD and EOD (lower = fairer). Best per attack-metric in bold among non-collapsed models; <sup>†</sup> marks near-chance models whose low disparity is an artefact of constant predictions.

| Aggregator                | Gaussian       |                    |                    | ALIE           |                  |                  | Fair-poison    |                  |                  |
|---------------------------|----------------|--------------------|--------------------|----------------|------------------|------------------|----------------|------------------|------------------|
|                           | AUC $\uparrow$ | DPD $\downarrow$   | EOD $\downarrow$   | AUC $\uparrow$ | DPD $\downarrow$ | EOD $\downarrow$ | AUC $\uparrow$ | DPD $\downarrow$ | EOD $\downarrow$ |
| fedavg                    | 0.511          | 0.031 <sup>†</sup> | 0.048 <sup>†</sup> | 0.935          | 0.062            | 0.039            | 0.945          | <b>0.015</b>     | 0.037            |
| bfwa                      | 0.483          | 0.014 <sup>†</sup> | 0.006 <sup>†</sup> | <b>0.991</b>   | 0.086            | 0.030            | 0.689          | 0.086            | 0.126            |
| krum                      | 0.986          | <b>0.053</b>       | 0.022              | 0.710          | <b>0.032</b>     | <b>0.003</b>     | <b>0.991</b>   | 0.049            | 0.007            |
| multikrum                 | 0.985          | 0.060              | <b>0.017</b>       | 0.916          | 0.063            | 0.030            | 0.989          | 0.068            | 0.024            |
| median                    | 0.962          | 0.066              | 0.033              | 0.790          | 0.051            | 0.020            | 0.980          | 0.038            | 0.010            |
| trimmed_mean              | 0.984          | 0.055              | 0.018              | 0.956          | 0.060            | 0.035            | 0.983          | 0.049            | <b>0.005</b>     |
| <b>robust_bfwa (ours)</b> | <b>0.992</b>   | 0.075              | 0.018              | 0.953          | 0.045            | 0.034            | 0.990          | 0.068            | 0.024            |

**Table 10:** Million-node scaling study (ogbn-products, 2.4M nodes, 123M edges,  $K=3$  clients, 15 rounds, CPU). \*2026 baseline. <sup>†</sup>DPD/EOD is a spurious artefact of a collapsed (near-chance) classifier, not a genuine fairness result—see discussion above. The energy column is a transparent *proxy*, not a metered measurement: logged wall-clock  $\times$  an assumed 65 W CPU draw (`src/trust/sustainability.py`), valid for relative comparison between methods on this one machine and not comparable across hardware.

| Method              | AUC $\uparrow$ | DPD $\downarrow$   | EOD $\downarrow$   | Time/run (s) | Energy (Wh) |
|---------------------|----------------|--------------------|--------------------|--------------|-------------|
| FedAvg-GAT          | 0.877          | 0.101              | 0.039              | 1449         | 26.2        |
| FairSIN             | <b>0.894</b>   | <b>0.095</b>       | <b>0.011</b>       | 1345         | 24.3        |
| FaVGNN* (2026)      | 0.495          | 0.000 <sup>†</sup> | 0.000 <sup>†</sup> | 2791         | 50.4        |
| DP-FedAvg           | 0.556          | 0.009 <sup>†</sup> | 0.021 <sup>†</sup> | 1251         | 22.6        |
| TrustFedGNN (no DP) | 0.861          | 0.114              | 0.071              | 3550         | 64.1        |
| <b>TrustFedGNN</b>  | 0.862          | 0.103              | 0.053              | 3665         | 66.2        |

**Table 11:** Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, and per-round communication. Energy is *not* reported here: wall-clock was not logged for these canonical small-graph runs, so the proxy of Section ?? has no input for them. We report it instead where the measurement exists, on the large-scale study (Table 10), rather than print an uninformative zero.

| Method              | ECE   | Brier | U-gap | Comm/rd (MB) | Trust |
|---------------------|-------|-------|-------|--------------|-------|
| FedAvg-GAT          | 0.031 | 0.054 | 0.002 | 0.225        | 0.650 |
| DP-FedAvg           | 0.322 | 0.274 | 0.002 | 0.215        | 0.480 |
| TrustFedGNN (no DP) | 0.043 | 0.061 | 0.002 | 0.896        | 0.632 |
| <b>TrustFedGNN</b>  | 0.062 | 0.073 | 0.006 | 0.896        | 0.718 |

**Table 12:** Composite trust score on Bail: utility, fairness, and privacy combined via the arithmetic-mean ( $p=1$ ) power mean, from the same canonical runs as Tables 4–6. Non-private methods score 0 on privacy by construction. The calibration-augmented composite is in Table 11. Ranking is sensitive to the reference constants (stated in text); best in bold.

| Method              | AUC   | DPD   | EOD   | $\epsilon$ | Trust        |
|---------------------|-------|-------|-------|------------|--------------|
| FedAvg-GAT          | 0.972 | 0.070 | 0.024 | –          | 0.579        |
| FairGNN             | 0.906 | 0.047 | 0.017 | –          | 0.582        |
| FairSIN             | 0.956 | 0.065 | 0.025 | –          | 0.577        |
| FairFed             | 0.961 | 0.054 | 0.018 | –          | 0.593        |
| F <sup>2</sup> GNN  | 0.971 | 0.064 | 0.021 | –          | 0.586        |
| DP-FedAvg           | 0.587 | 0.001 | 0.012 | 8          | 0.685        |
| TrustFedGNN (no DP) | 0.990 | 0.070 | 0.015 | –          | 0.593        |
| <b>TrustFedGNN</b>  | 0.988 | 0.066 | 0.009 | 8          | <b>0.767</b> |

**Table 13:** Mapping of TrustFedGNN components to EU AI Act articles and NIST AI RMF functions.

| Requirement                                    | Addressed by                                                                                                                                  |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <i>EU AI Act</i>                               |                                                                                                                                               |
| Art. 10 - Data & data governance               | Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.    |
| Art. 13 - Transparency & information           | Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics. |
| Art. 14 - Human oversight                      | Explicit, human-set fairness budget tau and privacy budget epsilon; Pareto trade-off curves let an operator choose the operating point.       |
| Art. 15 - Accuracy, robustness & cybersecurity | Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.          |
| <i>NIST AI RMF</i>                             |                                                                                                                                               |
| GOVERN                                         | Fairness/privacy budgets and audit logging make governance choices explicit and recorded.                                                     |
| MAP                                            | Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.                                                  |
| MEASURE                                        | DPD/EOD/EO, RDP epsilon, attack-retention, ECE and a composite trust score are all measured.                                                  |
| MANAGE                                         | Robust aggregation + fairness constraint actively mitigate measured risks each round.                                                         |

**Table 14:** Hyperparameters. “Explored” gives the sweep range where applicable.

| Hyperparameter             | Default                   | Explored                       |
|----------------------------|---------------------------|--------------------------------|
| GNN layers $L$             | 2                         | $\{2, 3\}$                     |
| Hidden dimension $d_h$     | 64                        | $\{32, 64, 128\}$              |
| Attention heads (FSER/GAT) | 4                         | $\{1, 4\}$                     |
| Dropout                    | 0.3                       | $\{0.2, 0.3, 0.5\}$            |
| Local optimiser            | AdamW                     | —                              |
| Local learning rate $\eta$ | 0.01                      | $\{10^{-3}, 10^{-2}\}$         |
| Weight decay               | $10^{-5}$                 | —                              |
| Local epochs $E$           | 2                         | $\{1, 2, 3\}$                  |
| Communication rounds $T$   | 40–60                     | per dataset                    |
| Clients $K$                | 3–10                      | $\{3, 5, 10, 20\}$             |
| Partition                  | Dirichlet( $\alpha=0.5$ ) | $\alpha \in \{0.1, 0.5, 1.0\}$ |
| Fairness weight $\lambda$  | 1.0                       | $\{0, 0.25, 0.5, 1, 2, 4\}$    |
| Fairness budget $\tau$     | 0.05                      | $\{0.01, 0.05, 0.1\}$          |
| DP $\epsilon$              | 8                         | $\{0.5, 1, 2, 4, 8, 16\}$      |
| DP $\delta$                | $10^{-5}$                 | —                              |
| DP clip $C$                | 1.0                       | —                              |
| Frank–Wolfe iters $T_{FW}$ | 20                        | —                              |
| Dual step $\eta_\mu$       | 0.1                       | —                              |
| MC-dropout samples         | 20                        | —                              |

**Table 15:** What each 2025–2026 baseline’s reimplementation keeps vs. drops.

| Baseline      | Kept                                                                                                   | Dropped / approximated                                                                                                                                                       |
|---------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FairGFL       | weight $\propto 1/(1+O_i)$ aggregation rule                                                            | $O_i$ (multi-graph node/edge overlap) proxied by sample-count deviation from the mean, since our setting has one Dirichlet-partitioned graph, not separate per-client graphs |
| FedGraph-Fair | minimax/DRO dual $\lambda$ reweighting high-loss clients                                               | personalised per-client models mixed via a learned peer-similarity graph (orthogonal decentralised-communication contribution)                                               |
| PUFFLE        | momentum controller auto-tuning the fairness weight toward a target disparity, + DP-SGD                | the third privacy channel (noised group-count sharing for clients missing a demographic group) – unneeded here since every client observes both groups                       |
| FedFACT       | closed-form global+local offset (exact reduction of the paper’s optimum for binary demographic parity) | general multiclass cost-matrix calibration and the iterative dual-ascent solver                                                                                              |
| PoPETs’25     | degree-2-polynomial FairFed weighting (the FHE-friendly surrogate itself)                              | threshold-CKKS secure aggregation and its noise-as-DP analysis (cryptographic infrastructure, no numeric effect once computed in the clear)                                  |